Module 1 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 1, Quiz B: Concept check

Reading: Module 1 Read 2 (complete Read 1 + Quiz A first)
Canvas: Concept Check quiz (Module 1)

Apply rules from Read 1–2: counting, conditional probability, combinations, and binomial process identification. **R and ggplot items** are on the Synthesis bank after the demo lab.

Excluded from Module 1: Bayes' rule — optional enrichment only (not on the course path).

Section A — Equally likely calculations (Read 1 §2)

A1. Two coin tosses

Fair coin tossed **twice**:

1. List S so outcomes are equally likely.
2. Let A = "at least one tail." List outcomes in A.
3. Find $P(A)$.

A2. Dice sum 11

Fair die rolled **twice**:

1. How many outcomes in S?
2. List outcomes with sum 11.
3. Find $P(\text{sum is } 11)$.

A3. Three coin tosses

Three fair tosses: find $P(\text{at least one head})$.

A4. Random babies (optional enrichment)

Three babies returned randomly to three mothers (all pairings equally likely):

1. How many outcomes in S?
2. $P(\text{all three correct})$.
3. $P(\text{none correct})$.

Answer key — Section A — Equally likely calculations (Read 1 §2)

1. $S = \{\text{HH,HT,TH,TT}\}$; 3 outcomes; $P = 3/4$.
2. $|S| = 36$; (5,6) and (6,5); $P = 2/36 = 1/18$.
3. $1 - P(\text{no heads}) = 1 - 1/8 = 7/8$.
4. List all assignments (A,B,C), (A,C,B), (B,A,C), (B,C,A), (C,A,B), (C,B,A): $|S| = 6$. $P(\text{all}) = 1/6$; $P(\text{none}) = 2/6 = 1/3$.

Section B — Conditional probability in context (Read 1 §4)

B1. Candy jar

10 candies: 6 red, 4 blue. Two drawn **without replacement**. A = first red, B = second red.

1. Find $P(A)$.
2. Find $P(B \mid A)$.
3. Use the multiplication rule for $P(A \text{ and } B)$.

B2. Two marbles (blue then green)

8 marbles: 4 blue, 4 green. Draw two **without replacement**. Find $P(\text{first blue and second green})$.

B3. Tiles: independence check

6 tiles: 2 red, 2 blue, 2 green; 3 round and 3 square (one of each color-shape). One tile drawn. Is "red" independent of "round"? Show reasoning.

B4. Two coin tosses

Fair coin tossed twice: find $P(\text{H on first and T on second})$.

B5. Conditional marbles

A bag has 8 marbles (4 blue, 4 green). Two marbles are drawn **without replacement**. Given the **first** marble is blue, find $P(\text{second is green})$.

B6. Two die rolls

Fair die rolled twice. A = first roll even, B = second roll is 6.

1. Are A and B independent? Explain briefly.
2. Compute $P(A \text{ and } B)$.

Answer key — Section B — Conditional probability in context (Read 1 §4)

1. $P(A) = 6/10$; $P(B \mid A) = 5/9$; $P(A \text{ and } B) = (6/10)(5/9) = 1/3$.
2. $(4/8)(4/7) = 1/7$.
3. Yes. $P(\text{red}) = 2/6 = 1/3$, $P(\text{red} \mid \text{round}) = 1/3$ (one red among three round tiles).
4. 1/4.
5. 4/7.
6. Yes (separate rolls); $P(A \text{ and } B) = P(A)P(B) = (1/2)(1/6) = 1/12$.

Section C — Two-way tables and independence (Read 1 §4 enrichment)

C1. COVID marginal and conditional

1000 people tested: 20 have COVID, 980 do not. Among COVID+: 17 test positive. Among COVID−: 78 test positive. Let C = has COVID, T^+ = tests positive.

1. Find $P(C)$ and $P(\text{not } C)$.
2. Find $P(T^+ \mid C)$ (sensitivity).
3. Find $P(T^+ \mid \text{not } C)$ (false positive rate).

C2. COVID joint probabilities

COVID testing study: 1000 people tested. 20 have COVID, 980 do not. Among COVID+: 17 test positive. Among COVID−: 78 test positive. Let C = has COVID, T^+ = tests positive.

1. Find $P(C \text{ and } T^+)$.
2. Find $P(\text{not } C \text{ and } T^+)$.

C3. COVID independence

1. If C and T^+ were independent, what would $P(C \text{ and } T^+)$ equal?
2. Are they actually independent? Explain using the definition.

C4. COVID overall positive rate

Find $P(T^+)$ — the overall probability of a positive test.

C5. Interpret $P(T^+)$

In one sentence, interpret $P(T^+)$ in the context of this COVID study.

Answer key — Section C — Two-way tables and independence (Read 1 §4 enrichment)

1. $P(C) = 20/1000 = 0.02$; $P(\text{not } C) = 0.98$; $P(T^+ \mid C) = 17/20 = 0.85$; $P(T^+ \mid \text{not } C) = 78/980 \approx 0.0796$.
2. $17/1000 = 0.017$; $78/1000 = 0.078$.
3. $P(C)P(T^+) = (0.02)(0.095) = 0.0019$; actual joint is 0.017 — **not** independent.
4. $(17 + 78)/1000 = 95/1000 = 0.095$.
5. Among all 1000 people tested, 9.5% received a positive test result.

Section D — Counting and combinations (Read 1 §5)

D1. Factorials

1. Compute 4!.
2. What is 0! by convention?

D2. Permutations and combinations

Compute ${}_5P_2$ and ${}_5C_2$.

D3. Order matters vs does not

1. Choose 3 from 7 for a **committee** (order does not matter): write the expression and compute.
2. Choose 3 from 7 for **three distinct roles**: write the expression and compute.

D4. Team from players

Choose 4 players from 10 for a team (order does not matter):

1. Write the expression using ${}_nC_r$ or $\binom{n}{r}$.
2. Compute the value.

Answer key — Section D — Counting and combinations (Read 1 §5)

1. $4! = 24$; $0! = 1$.
2. ${}_5P_2 = 20$; ${}_5C_2 = 10$.
3. $\binom{7}{3} = 35$; ${}_7P_3 = 210$.
4. $\binom{10}{4} = 210$.

Section E — Binomial random process (Read 1 §6–7)

E1. Four BRP conditions

State the **four conditions** for a binomial random process.

E2. Coin flips

Fair coin flipped 12 times, X = number of heads.

1. What is n?
2. What does x mean in $P(X = x)$?
3. What is p in this context?

E3. Germination

20% of seeds fail; plant 8 seeds, X = number that fail.

1. Identify n and p.
2. Write an expression for $P(X = 3)$ (no simplification required).

Answer key — Section E — Binomial random process (Read 1 §6–7)

1. Binary outcomes; fixed n; fixed p each trial; independent trials.
2. $n = 12$ trials; x is a possible count of heads; $p = 0.5$ = success prob. on one toss.
3. $n = 8$, $p = 0.20$; $P(X = 3) = \binom{8}{3}(0.2)^3(0.8)^5$.
